Introduction to Economic and Social Network

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Thanks to typst!

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1. Games on networks

1.1. Games on networks

Consider the following game:

- Each player chooses $x_i = 0, 1$
- Payoff depends on
  - how many neighbors choose each action
  - how many neighbors a player has

Consider cases where i's payoff is

$$u_{d_i}\left(x_i, m_{N_i}\right) \tag{1}$$

where m_{N_i} is the number of neighbors of i choosing 1 and $d_i(g)$ is the number of neighbors of i.

Let N_i be the set of i's neighbors. Then $d_i(g) = |N_i|$ and

$$m_{N_i} = \#\left\{j \in N_i, x_j = 1\right\} \tag{2}$$

Example 1.1.1 (Simple Complement): Agent i is willing to choose 1 $\iff$ at least t neighbors do:

- payoff action 0: $u_{d_i}\left(0, m_{N_i}\right) = 0$
- payoff action 1: $u_{d_i}\left(1, m_{N_i}\right) = -t + m_{N_i}$

In Example 1.1.1, the payoff difference is

$$u_{d_i}\left(1, m_{N_i}\right) - u_{d_i}\left(0, m_{N_i}\right) = -t + m_{N_i} \tag{3}$$

. The best response is:

$$x_i^*(m) = \begin{cases} 1 \text{ if } m \geq t \\ 0 \text{ o.w.} \end{cases} \tag{4}$$

Now let $t = 2$:

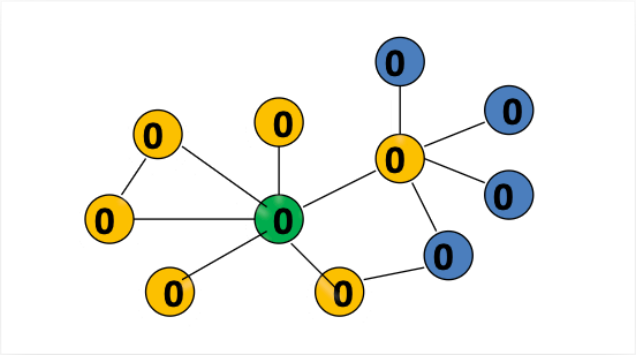


Figure 1: Equilibrium 1: all choose 0

Another equilibrium is shown in Figure 2:

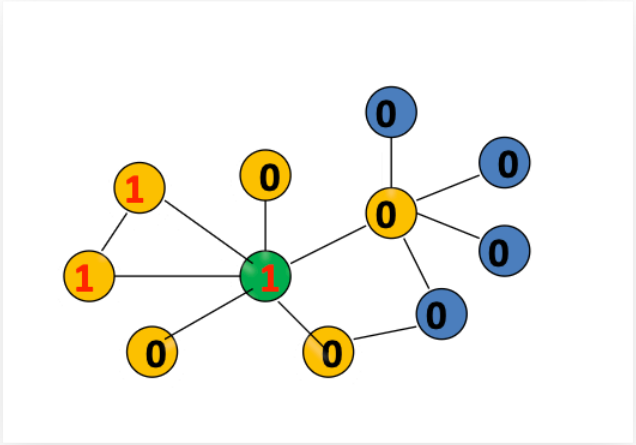


Figure 2: Equilibrium 2: 3 nodes choose 1

1.2. Complements and substitutes

Definition 1.2.1 (Externalities):
- Others' behaviors affect my utility/welfare
- Others' behaviors affect my decisions, actions, consumptions, opinions...

Others' actions affect the relative payoffs to my actions.

1.2.1. Complements

Choice to take an action by my friends increases my relative payoff to taking that action (e.g., friend learns to play a video game)

Let $m = m_{N_i}$ as defined in Equation 2, then strategic complements means:

Definition 1.2.2 (Complements): for all d[1] and $m \geq m'$, we have

$$u_d(1,m) - u_d(0,m) \geq u_d(1,m') - u_d(0,m') \tag{5}$$

1.2.2. Substitutes

Choice to take an action by my friends decreases my relative payoff to taking that action (e.g., roommate buys a stereo/fridge)

Strategic substitutes satisfy

Definition 1.2.3 (Substitutes): for all $d \wedge m \geq m'$:

$$u_d(1,m) - u_d(0,m) \leq u_d(1,m') - u_d(0,m') \tag{6}$$

Definition 1.2.4 (Nash Eq.): Every player's action is optimal for that player given the actions of others

We often look for pure strategy equilibria and may require some mixing to get existence (hopefully not...)

1.3. Strategic complements

Example 1.3.1 (Examples):

- education decisions
- care about number of neighbors, access to jobs, etc.: invest if at least k neighbors do
- smoking & other behavior among teens, peers, ...
- technology adoption: how many others are compatible
- learn a language, ...
- cheating, doping

One useful observation is that complements $\implies$ there exists a threshold $t(d)$ s.t. i prefers

$$\text{BR}_i = \begin{cases} 1 \text{ if } & m_{N_i} > t(d) \\ 0 \text{ if } & m_{N_i} < t(d) \end{cases} \tag{7}$$

and can be indifferent at the threshold $t(d)$.

[1] For a given agent i, we suppress i in the subscript.

Consider a case where t doesn't depend on d: $t \equiv 2$, then there are multiple equilibria as in Figure 3.

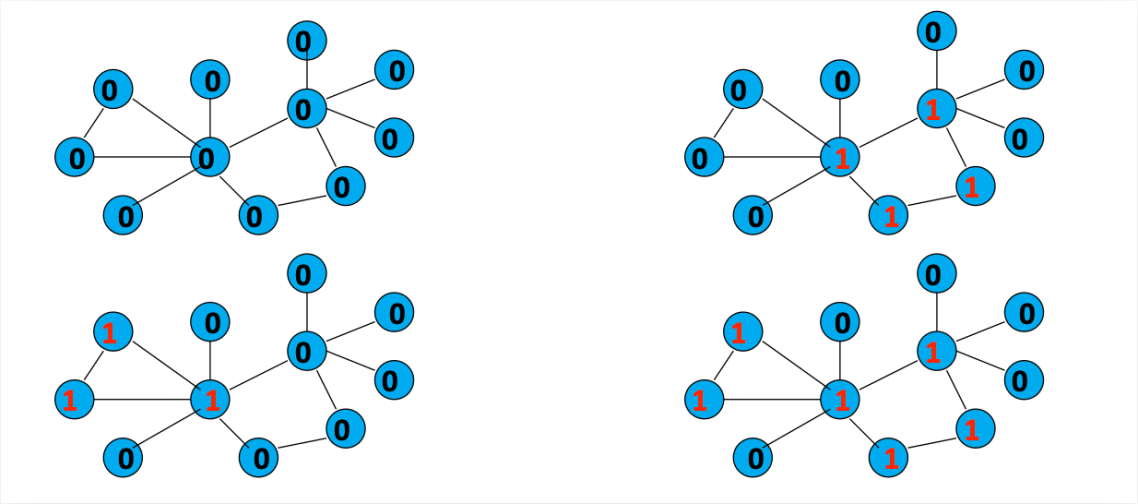


Figure 3: Multiple Equilibria

1.3.1. Lattice structure

The equilibria in Figure 3 have a *lattice structure*.

Definition 1.3.1 (Complete Lattice): A partially ordered set X is a complete lattice if every subset $A \subseteq X$ has a least upper bound $\sup_X A$ and a greatest lower bound $\inf_X A$ in X. This includes the empty set: $\sup_X \emptyset = \bot$ and $\inf_X \emptyset = \top$.

Definition 1.3.2 (Coordinate order): For 2 action allocation: $x = (x_i), y = (y_i)$, we define $x \leq y$ as

$$x \leq y \iff x_i \leq y_i, \forall i = 1, 2, ..., n \tag{8}$$

When the equilibrium set is a complete lattice under Definition 1.3.2, it has a unique least and greatest equilibrium. Joins and meets are taken *within the equilibrium set*; coordinate-wise maxima or minima of equilibria need not be equilibria.

Proposition 1.3.1: Consider a finite-player game. Each $S_i \subset \mathbb{R}^{k_i}$ is a nonempty compact sublattice under coordinate order. Each payoff is continuous, supermodular in the player's own action, and has increasing differences in (x_i, x_{-i}). Then the pure-strategy equilibrium set is a nonempty complete lattice. (Zhou, 1994)

Proposition 1.3.1 implies:

1. Pure-strategy equilibrium exists.
2. There is a least and a greatest equilibrium.
3. In the binary threshold games above, least best responses iterated from all 0 find the least equilibrium; greatest best responses from all 1 find the greatest.

1.3.2. Action spaces and best responses

For a rectangular action space $S_i = \prod_{\ell=1}^{k_i} [a_\ell, b_\ell]$, joins and meets of nonempty subsets are coordinate-wise suprema and infima. Its least and greatest elements are $\bot_i = (a_\ell)_\ell$ and $\top_i = (b_\ell)_\ell$. For every $A \subseteq S_i$, $\sup_{S_i} A \leq \top_i$; equality holds whenever A contains $\top_i$, including proper subsets.

Supermodularity makes own action coordinates complements (automatic for a scalar action). Increasing differences means that higher opponents' actions raise the return to increasing one's own action. The compactness and continuity conditions in Proposition 1.3.1 ensure best responses exist.

1.4. An application in labor

Dropout decisions as in (Calvó-Armengol & Jackson, 2004; 2007). Value to being in the labor market depends on # of friends in the market

- Dropout if some # of friends do so
- Some heterogeneity in threshold (different costs, natural abilities...)
- Homophily: segregation in the network

Consider this cascade model as in Figure 4, first, 2 groups exhibit homophily, and secondly, *one agent dropout if at least half of its neighbors do so*.

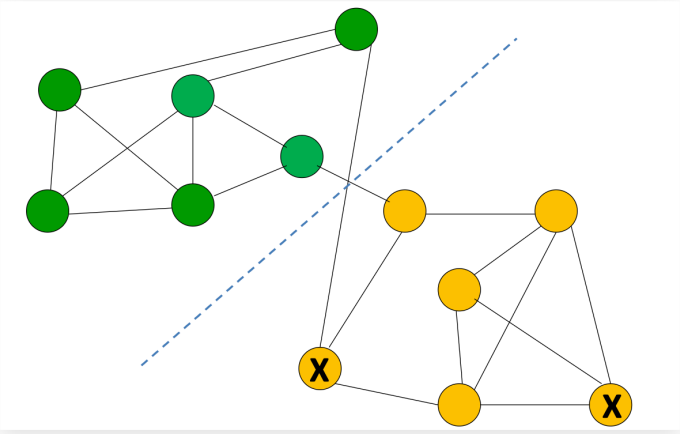


Figure 4: Cascade model

As in Figure 4, there are 2 initial dropouts. The cascade ends at Figure 5, where the entire yellow component drops out.

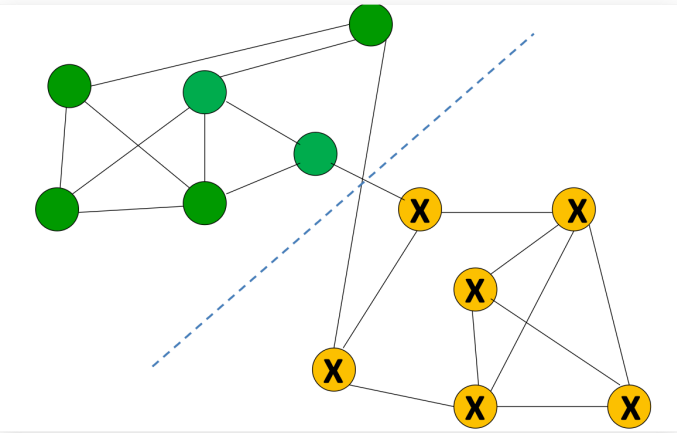


Figure 5: Cascade model: result

1.5. Strategic substitutes

Similarly, consider the following Example 1.5.1:

Example 1.5.1:
- information gathering
  - e.g., payoff of 1 if anyone in neighborhood is informed, cost to being informed ($c < 1$)
- local public goods (shareable products...)
- competing firms (oligopoly with local markets)

And similar to Equation 7, we have the threshold function $t(d)$ s.t.

$$\text{BR}_i = \begin{cases} 1 \text{ if } m_{N_i} < t(d) \\ 0 \text{ if } m_{N_i} > t(d) \end{cases} \tag{9}$$

and can be indifferent at $m_{N_i} = t(d)$.

Example 1.5.2 (Best Shot Game): One is willing to choose 1 $\iff$ no neighbors do. Each node i chooses action $x_i \in \{0, 1\}$ where $x_i = 1$ means providing a public good at cost c. The payoff is

$$u_{d_i}\left(0, m_{N_i}\right) = \begin{cases} 1 \text{ if } m_{N_i} > 0 \\ 0 \text{ o.w.} \end{cases}, u_{d_i}\left(1, m_{N_i}\right) = 1 - c \tag{10}$$

and let $c \in (0, 1)$.

Then for Example 1.5.2, suppose a network in form of Figure 6,

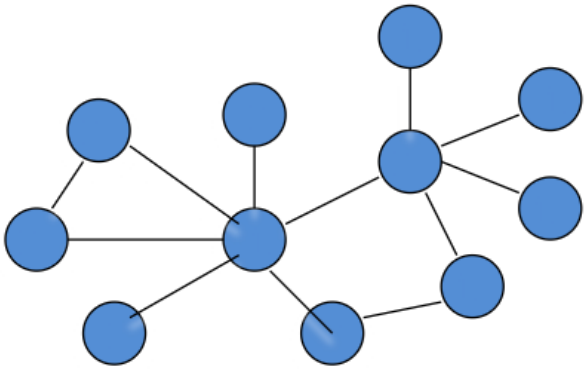


Figure 6: Network structure for Example 1.5.2

It's clear that one will choose $x_i = 1$ if and only if $m_{N_i} = 0$. One eq. is described in Figure 7.

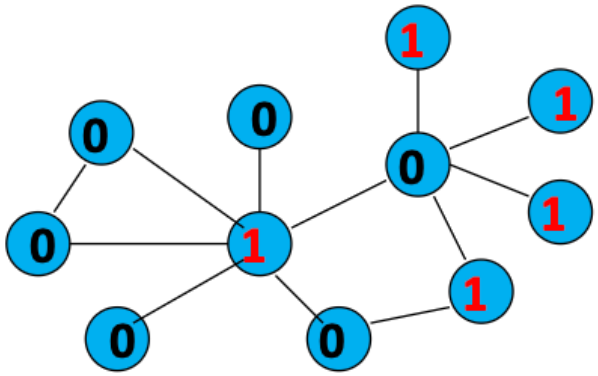


Figure 7: An eq. of Example 1.5.2

Other equilibria are possible.

Consider the best shot game in Example 1.5.2 for a star network, then we have only 2 eq. as in Figure 8. The equilibria have *different utility distributions and total costs*. The eq. is related to a concept called **maximal independent set**.

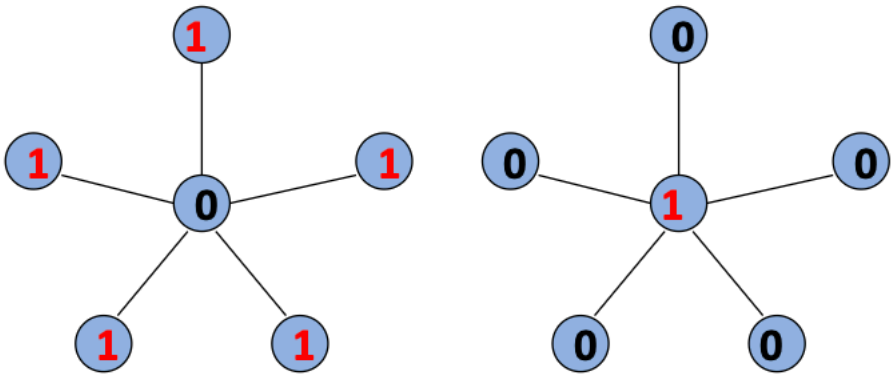


Figure 8: eq. of star network

Definition 1.5.1 (Maximal Independent Set): Let $M \subseteq N$ be the set of providers:

$$M \equiv \{i \in N, x_i = 1\} \tag{11}$$

1. independent: for all $i, j \in M$, ij $\notin g$
2. maximal: for all $k \notin M$, if we add k to $M \cup \{k\}$, then $M \cup \{k\}$ is not *independent*.

1.6. Summary

In a game of strategic **complements**:

1. Under the hypotheses of Proposition 1.3.1, pure-strategy equilibria form a nonempty complete lattice.

In a game of strategic **substitutes**:

1. Best shot game: pure strategy equilibria exist and are related to maximal independent sets
2. Others: pure strategy may not exist, but mixed will (with finite action spaces)
3. Equilibria usually do not form a lattice

2. TA session 1

2.1. Nodes and networks

Definition 2.1.1 (Nodes): set

$$N = \{1, 2, ..., n\} \tag{12}$$

Definition 2.1.2 (Network): A network (N, g) consists of nodes N and an $n \times n$ adjacency matrix g where:

- $g_{ij} = 1 \implies i$ and j are linked
- A network is undirected iff $g_{ij} = g_{ji}$
- Usually $g_{ii} = 0$

g can also be a collection of links, say, $g = \{12, 23\}$, then $(ij) \in g$ means i and j are linked.

2.2. Neighborhood and degree

Definition 2.2.1 (Neighborhood of i): The set

$$N_i(g) \equiv \left\{ j \in N : g_{ij} = 1 \right\} \tag{13}$$

similarly, degree is then defined as

Definition 2.2.2 (Degree):

$$d_i(g) = \#N_i(g) \tag{14}$$

2.3. Connecting nodes

Definition 2.3.1 (Walk): A walk from i to j is a sequence of nodes $\{i_t\}_{t=1}^{K}$ and sequence of links $\{i_t i_{t+1}\}_{t=1}^{K-1}$ such that

$$i_t i_{t+1} \in g, \forall k \in \{1, 2, ..., K-1\} \tag{15}$$

and $i_1 = i, i_K = j$.

Definition 2.3.2 (Path): A path is a walk with each node in $\{i_t\}_{t=1}^K$ is **distinct.**

Definition 2.3.3 (Cycle): A cycle is a walk $\{i_t\}_{t=1}^K$ w/
1. $i_1 = i_K$
2. all other nodes are distinct

Definition 2.3.4 (Geodesic): The shortest path between 2 nodes.[2]

2.4. Component

Definition 2.4.1 (Connected): A network (N, g) is connected if there is a **Path** between every 2 nodes.

Definition 2.4.2 (Component): A non-empty subnetwork (N', g') such that:$N' \subseteq N, g' \subseteq g, N' \neq \emptyset$ and
- (N', g') is connected
- if $i \in N'$ and $ij \in g$, then $j \in N'$ and $ij \in g'$

Components are the maximal connected subgraphs of a network.

Definition 2.4.3 (Complete network): All possible links are present:

$$g = \mathbf{1}\mathbf{1}^\top - I \tag{16}$$

or equivalently, $g_{ij} = 1$ for all $i \neq j$.

2.5. Diameter and average path length

Definition 2.5.1 (Diameter): The largest geodesic as defined in Definition 2.3.4 between any 2 nodes in the network.

[2] If there is no path between 2 nodes, then the distance between them is ∞.

For a disconnected network, the convention in footnote[2] gives diameter ∞. In practice, we usually report the diameter of the giant component.

For a balanced d-ary tree with $d \geq 2$, height and diameter grow logarithmically with the number of nodes:

$$D \propto \frac{\log(n)}{\log(d)} \tag{17}$$

A general tree can have diameter $n - 1$, as a path does.

Definition 2.5.2 (Average path length): The average geodesic between any 2 nodes in the network, or formally,

$$\sum_{i \geq j} \ell(i,j) \left[\frac{n(n-1)}{2} \right]^{-1} \tag{18}$$

2.6. Centrality measures

Centrality measures the importance of a node. Degree centrality measures *how connected a node is*.

Definition 2.6.1 (Degree Centrality): Let $d_i(g)$ be the degree of node i where

$$d_i(g) = \#\left\{ j \in N, g_{ij} = 1 \right\} \tag{19}$$

or in the normalized form:

$$\frac{1}{n-1} d_i(g) \tag{20}$$

Closeness centrality uses shortest-path distances:

Definition 2.6.2 (Closeness Centrality): Let

$$C_i(g) = \frac{n-1}{\sum_{j=1}^{n} l(i,j)} \tag{21}$$

where $l(i,j)$ is the number of links in the shortest path between i and j.

Definition 2.6.3 (Decay Centrality): Let

$$C_i^d(g) = \sum_{j=1, j\neq i}^{n} \sigma^{l(i,j)} \tag{22}$$

where $l(i,j)$ is defined similar to that of Definition 2.6.2. Its normalized form is

$$\frac{1}{(n-1)\sigma} C_i^d(g) \tag{23}$$

For $n > 1$, take $0 < \sigma < 1$; unreachable nodes contribute zero. If i's component has m_i nodes:

- As $\sigma \to 1$, $C_i^d(g) \to m_i - 1$, the number of *other* nodes in the component.
- As $\sigma \to 0$, the raw $C_i^d(g)$ tends to zero, while $\frac{C_i^d(g)}{(n-1)\sigma} \to \frac{d_i(g)}{n-1}$. Only distance-one terms survive this normalization.
- Intermediate values discount distant nodes.

Betweenness centrality measures *how much a node connects other nodes*.

Definition 2.6.4 (Betweenness Centrality): For $n \geq 3$, let

$$C_k^B(g) = \frac{\sum_{i<j, i\neq k, j\neq k} \frac{P_k(ij)}{P(ij)}}{\binom{n-1}{2}} = \frac{2\sum_{i<j, i\neq k, j\neq k} \frac{P_k(ij)}{P(ij)}}{(n-1)(n-2)} \tag{24}$$

be the betweenness centrality of node k[3], where

1. $P_k(ij)$ denotes the total number of geodesics between i, j that k lies on.
2. $P(ij)$ is the total number of geodesics between i, j. A disconnected pair contributes zero.

2.7. Prestige, influence, eigenvector-based centrality

Eigenvector centrality weights connections by the centrality of neighboring nodes.

[3]The normalization $\binom{n-1}{2}$ counts pairs among the other $n-1$ nodes, whose geodesics may pass through this node.

Definition 2.7.1 (Eigenvector Centrality): Given a network (N, g), we define the eigenvector centrality of a node i as:

$$C_i^e(g) = \alpha \sum_{j \in N} g_{ij} C_j^e(g) \tag{25}$$

and in vector form:

$$C^e = \alpha g C^e \tag{26}$$

where $C^e(g) \in \mathbb{R}^n$.

Rearrange Equation 26 as $\frac{1}{\alpha} C^e = g C^e$. Setting $\lambda = \frac{1}{\alpha}$ gives the eigenvalue equation for g. We seek the eigenvector associated with the **largest** eigenvalue, which is nonnegative.

Katz-Bonacich centrality sums the contributions from direct and indirect connections.

2.7.1. Number of walks from i to j

Consider a network (N, g), then g^n indicates how many walks there are of length n between 2 nodes, see Example 2.7.1.

Example 2.7.1: Let $g^2 = \begin{pmatrix} 2 & 0 & 0 & 2 \\ 0 & 2 & 2 & 0 \\ 0 & 2 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{pmatrix}$, then $\left(g^2\right)_{14} = 2$ counts the two walks from node 1 to node 4 of length 2.

Give each node an initial value $\alpha d_i(g)$ for $\alpha > 0$. Assume $\beta \geq 0$ and $\beta \rho(g) < 1$, so the walk series converges.

1. one step adds $\alpha\beta \sum_{j \in N} g_{ij} d_j(g)$
2. two steps add $\alpha\beta^2 \sum_{j \in N} \left(g^2\right)_{ij} d_j(g)$
3. ...
4. summing over all lengths gives

$$\begin{aligned} C^b(g) &= \alpha g \mathbf{1} + \alpha\beta g^2 \mathbf{1} + \ldots \\ &= \alpha\left(I + \beta g + \beta^2 g^2 + \ldots\right) g \mathbf{1} \\ &= \alpha (I - \beta g)^{-1} g \mathbf{1} \end{aligned} \tag{27}$$

and note that $g\mathbf{1} \in \mathbb{R}^{n \times 1}$ is the degree of each node.

- β weights each additional step of a walk.
- α scales centrality.

2.8. Clustering

"If i has relationships with both j and k, how likely is it that j and k are related ?"

This problem is central to clustering. In sociology, this is known as a **triadic closure**: If A knows B and A knows C, then B and C are very likely to know each other as well.

2.8.1. Individual clustering coefficient

Definition 2.8.1 (Individual Clustering Coefficient): Let

$$Cl_i(g) = \frac{\#\{jk \in g | j,k \in N_i(g)\}}{\#\{jk \mid j,k \in N_i(g)\}} \tag{28}$$

be the Individual Clustering Coefficient of node i.

The number of neighbor pairs is

$$\#\{jk \mid j,k \in N_i(g)\} = \binom{d_i(g)}{2} \tag{29}$$

1. In the Erdős–Rényi network $G(n,p)$, links between neighbors are independent of the links to i. Conditional on a neighbor set with at least two nodes, $\mathbb{E}[Cl_i(g) \mid N_i(g)] = p$. Realized clustering fluctuates around this value.
2. If the degree of node i is $d_i(g) \leq 1$ (i.e., it has 0 or 1 links), then it has insufficient neighbors to form any pair of neighbors. Then

$$Cl_i(g) = 0 \tag{30}$$

2.8.2. Overall clustering

Definition 2.8.2 (Overall clustering): Let

$$Cl(g) = \frac{\sum_{i \in N} \#\{jk \in g | \, j,k \in N_i(g)\}}{\sum_{i \in N} \#\{jk \mid j,k \in N_i(g)\}} \tag{31}$$

be the overall clustering coef. of a network (N,g).

Each centered triple receives equal weight. A node with more neighbors contributes more triples, so overall clustering weights high-degree nodes more heavily than average clustering does.

2.8.3. Average clustering

Definition 2.8.3 (Average clustering): Let

$$Cl^{\text{avg}}(g) = \frac{1}{n}\sum_{i=1}^{n} Cl_i(g) \tag{32}$$

be the average clustering coefficient of a network (N, g).

Equal weight for every node.

3. Games on networks II

3.1. Strategic substitutes (Bramoullé & Kranton, 2007)

You need to acquire information for an application, the payoff is

$$f\left(x_i + \sum_{j \in N_i(g)} x_j\right) - cx_i \tag{33}$$

where f is a concave "production function", FOC gives

$$f'(x^*) = c \tag{34}$$

So in all pure strategy Nash Equilibria, we have the optimization:

$$\begin{aligned} \max_{x_i} \quad & f\left(x_i + \sum_{j \in N_i(g)} x_j\right) - cx_i \\ \text{s.t.} \quad & x_i \geq 0 \end{aligned} \tag{35}$$

then KKT condition of Equation 35 gives:

$$\begin{aligned} f'\left(x_i + \sum_{j \in N_i(g)} x_j\right) &= c - \lambda_i \\ \lambda_i &\geq 0 \\ x_i &\geq 0 \\ \lambda_i x_i &= 0 \end{aligned} \tag{36}$$

and that is

$$\begin{aligned} x_i + \sum_{j \in N_i(g)} x_j &\geq f'^{-1}(c) \\ x_i = 0 \text{ if } x_i + \sum_{j \in N_i(g)} x_j &> f'^{-1}(c) \end{aligned} \tag{37}$$

There are 2 types of equilibria:

- Distributed eq.

$$x_i \in \left(0, f'^{-1}(c)\right) \tag{38}$$

  for some i's
- Specialized eq. for each i, either $x_i = 0$ or $x_i = f'^{-1}(c)$

We consider the case where $x^* = 1$, that is, everyone wants a total investment of at least 1 in her neighborhood. The equilibria is described in Figure 9.

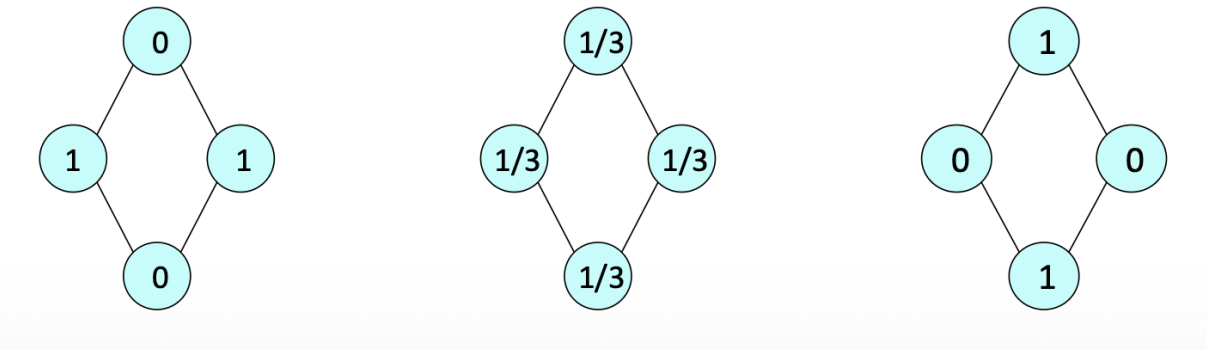


Figure 9: Eq. of BK 07

The specialized equilibria can be characterized using Definition 1.5.1 as follows:

Proposition 3.1.1: Specialized Nash equilibria are such that a maximal independent set = the specialist set

$$\{i : i \in N, x_i = f'^{-1}(c)\} \tag{39}$$

3.2. A linear quadratic model

This comes from (Ballester et al., 2006). Assume $a, b > 0$, $w_{ij} \geq 0$, $w_{ii} = 0$, and $x_i \geq 0$. The payoff is:

$$u(x_i, x_{-i}) = ax_i - 0.5bx_i^2 + \sum_j w_{ij}x_i x_j \tag{40}$$

and we set quadratic intrinsic value to be $v(x_i) = ax_i - 0.5bx_i^2$. This is also Strategic Complements since

$$\frac{\partial u}{\partial x_i} = a - bx_i + \sum_j w_{ij}x_j \tag{41}$$

so when x_j increases, $\frac{\partial u}{\partial x_i}$ also increases.

To solve the best response of x_i given x_{-i}, use the first-order condition:

$$a - bx_i + \sum_j w_{ij}x_j = 0 \tag{42}$$

so

$$x_i^* = \frac{1}{b}\left(a + \sum_j w_{ij}x_j\right) \tag{43}$$

consider the matrix form where $\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ and $\boldsymbol{\alpha} = \frac{a}{b}\mathbf{1}_n, \boldsymbol{g} = \left(g_{ij}\right)_{n\times n}$ where $g_{ij} = \frac{w_{ij}}{b}$, then we can rewrite Equation 42 in matrix form:

$$\boldsymbol{x} = \boldsymbol{\alpha} + \boldsymbol{g}\boldsymbol{x} \tag{44}$$

If $\rho(\boldsymbol{g}) < 1$, the Neumann series converges and gives

$$\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{g})^{-1}\boldsymbol{\alpha} = \sum_{k=0}^{\infty} \boldsymbol{g}^k \boldsymbol{\alpha}. \tag{45}$$

A unique finite Nash equilibrium exists when $\rho(g) < 1$, equivalently $b > \rho(W)$ for $W = \left(w_{ij}\right)$.

- w_{ij}: Strength of network complementarity. Large w strengthens the feedback loop.
- b: Curvature of the cost function (diminishing returns).
  - Large b makes marginal costs rise steeply and limits increases in x_i.

Recall Katz - Bonacich centrality:

$$B(g) = (I - g)^{-1} g\mathbf{1} = \sum_{k\geq 0} g^{k+1}\mathbf{1} \tag{46}$$

so the equilibrium is

$$\boldsymbol{x} = \frac{a}{b}(\mathbf{1} + B(g)). \tag{47}$$

3.3. Friendship paradox

Example 3.3.1 (Friendship Paradox): In an undirected network, write $F_i = \left(\frac{1}{d_i}\right)\sum_{j\in N_i} d_j$ when $d_i > 0$, and $F_i = 0$ for an isolated node. Compare the population mean degree $\mu = \left(\frac{1}{n}\right)\sum_i d_i$ with the average person's mean friend degree $\overline{F} = \left(\frac{1}{n}\right)\sum_i F_i$. Then $\overline{F} \geq \mu$.

Summing over each undirected edge once gives

$$\begin{aligned} \overline{F} - \mu &= \frac{1}{n} \sum_{i<j:g_{ij}=1} \left(\frac{d_i}{d_j} + \frac{d_j}{d_i} - 2 \right) \\ &= \frac{1}{n} \sum_{i<j:g_{ij}=1} \frac{\left(d_i - d_j\right)^2}{d_i d_j} \geq 0. \end{aligned} \tag{48}$$

The inequality is strict exactly when some edge joins nodes of different degrees. This is the argument in (Jackson, 2019), Lemma 1.

Sampling a uniformly random edge endpoint assigns node j probability $\frac{d_j}{\sum_k d_k}$. For this sampling rule, the mean degree is $\frac{\mathbb{E}[d^2]}{\mathbb{E}[d]} = \mu + \frac{\mathbb{V}[d]}{\mu}$ when $\mu > 0$. This generally differs from $\overline{F}$.

Example 3.3.2 (The Impact of the Friendship Paradox): Consider a 0-1 game...
- People choose solid or plaid
- Prefer to match the majority of your friends
- Go with own preference if indifferent

Start from the following state:

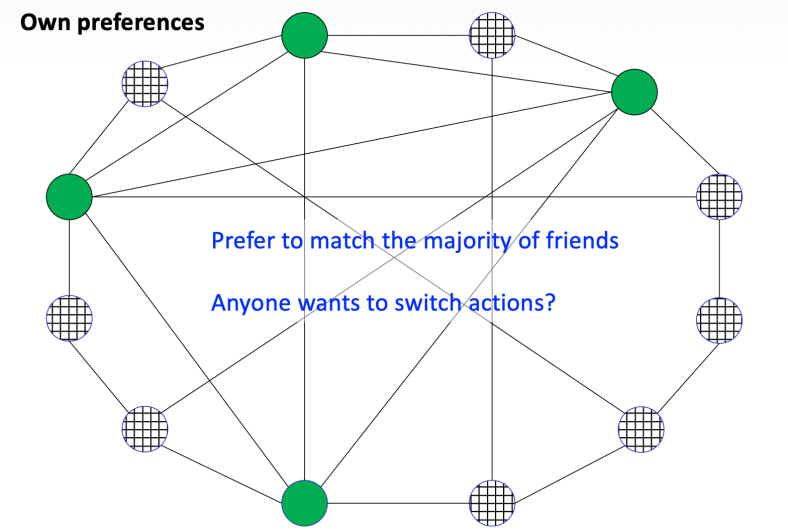


Figure 10: Own preferences

Note that in this starting state, only the influencers prefer the *solid* ones. Then each node tries to match the majority of its friends. The resulting equilibrium is:

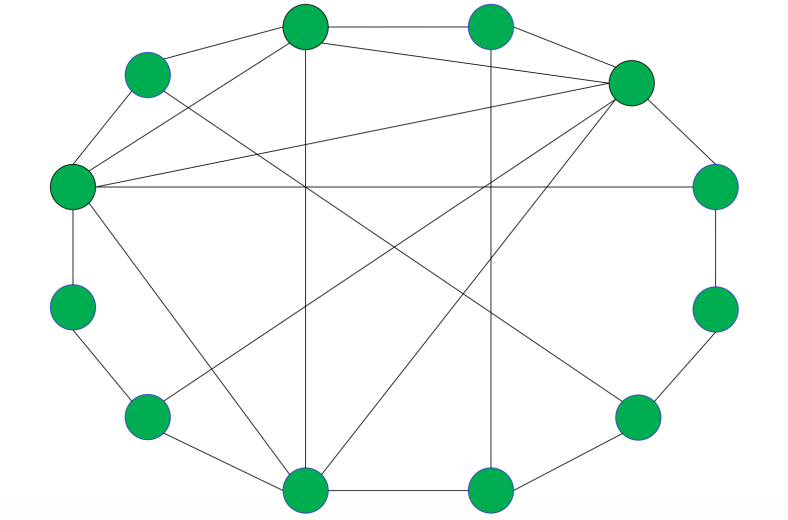


Figure 11: Eq. of this shirt game

That is, all nodes will choose solid in the end.

In this example of Example 3.3.1, the social average *preference for solid shirts* is $\frac{4}{12} = \frac{1}{3}$.

Each node's perceived average *preference for solid shirts* is shown below:

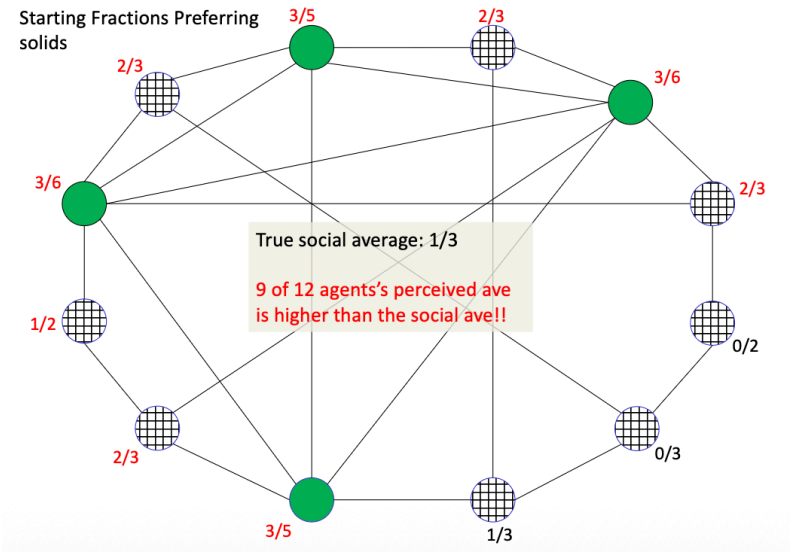


Figure 12: Perceived averages

For 9 of the 12 agents, the perceived average exceeds the social average.

The friendship paradox makes socially active agents overrepresented among observed friends. Their behavior can then appear more common than it is in the population, the *majority illusion*.

3.3.1. Empirical observations

1. Systematic Overestimation (Perkins et al., 2005)

TABLE 1. Students' accuracy in perceiving the drinking norm at their school (comparing actual with perceived number of alcoholic drinks consumed the last time students "partied"/socialized)

Actual school norm (median drinks)	Accuracy of perceived drinking norm: Underestimate of ≥3 drinks (%)	Underestimate of 1-2 drinks (%)	Accurate estimate (%)	Overestimate of 1-2 drinks (%)	Overestimate of ≥3 drinks (%)	Total	*N* of respondents	*N* of schools
0	NA	NA	20.6	19.5	59.9	=100%	1,891	4
1	NA	10.5	3.8	28.5	57.2	=100%	2,526	6
2	NA	7.5	8.1	30.6	53.7	=100%	8,345	14
3	3.8	6.4	13.5	37.5	38.7	=100%	18,859	35
4	3.1	12.3	12.6	37.0	34.9	=100%	20,353	38
5	4.3	15.8	20.6	24.1	35.3	=100%	11,481	20
6	6.9	23.2	15.0	23.5	31.5	=100%	8,912	12
7	5.7	23.3	9.7	23.6	37.8	=100%	352	1
Total schools	3.4	11.8	13.8	31.9	39.1	=100%	72,719	130

   The vast majority of people (31.9% + 39.1% ≈ 71%) seriously overestimated their classmates' alcohol consumption.
2. Positive Correlation(degree, behavior) For middle school students (12-15 years old)...
   1. More incoming friendship nominations were associated with smoking at follow-up. (Valente et al., 2005)

2. In (Tucker et al., 2013), five additional incoming friendship nominations were associated with a predicted next-wave past-month drinking probability of 10.4% rather than 8%, a 30% relative increase in the paper's example.

(Jackson, 2019) models how strategic complementarity and friendship formation can generate a positive association between degree and actions.

3.4. Friendship paradox in (Jackson, 2019)

Consider the following model:

- there are n players, each with intrinsic preference θ_i
- each one chooses an action x_i
- in a random network
- Let $P_i(d)$ denote the distribution of population of i's potential neighbors' degrees.

Lemma 3.4.1 (Expected Degree of Neighbors): Under degree-weighted neighbor sampling, assume $\mathbb{E}_i[d] > 0$. The neighbor degree distribution is $\frac{P_i(d)d}{\mathbb{E}_i[d]}$, so $\hat{\mathbb{E}}_i[d] = \frac{\mathbb{E}_i[d^2]}{\mathbb{E}_i[d]} = \mathbb{E}_i[d] + \frac{\mathbb{V}_i[d]}{\mathbb{E}_i[d]} \geq \mathbb{E}_i[d]$. The inequality is strict if and only if $\mathbb{V}_i[d] > 0$.

Proposition 3.4.1: Under the common-belief linear-quadratic model of Lemma 3.5.1, assume $\hat{\mathbb{E}}[\theta] = \mathbb{E}[\theta] > 0$ and $\mathbb{E}[d] > 0$. Then

$$x_i^{\text{prec}}(\theta_i, d_i) \geq x_i^{\text{soc}}(\theta_i, d_i), \forall \theta_i, d_i \tag{49}$$

and

$$\underbrace{\hat{\mathbb{E}}\left[x_i^{\text{prec}}(\theta_i, d_i)\right] \geq}_{\text{network perception effect}} \underbrace{\mathbb{E}\left[x_i^{\text{prec}}(\theta_i, d_i)\right] \geq \mathbb{E}[x_i^{\text{soc}}(\theta_i, d_i)]}_{\text{network eq. effect}} \tag{50}$$

In Equation 49, overestimating others' actions raises one's own action through strategic complementarity.

For Equation 50,

1. network perception effect: *You believe others are doing more than they actually are.*
2. network eq. effect: *Strategic complementarity allows you to do more in eq.*

3.5. Further results

3.5.1. More on (Bramoullé & Kranton, 2007)

1. Any maximal independent set is an eq.
2. How to select one?

- idea: Stability as Equilibrium Selection

A stable equilibrium is restored after small perturbations. A small perturbation of an unstable equilibrium can lead to a different equilibrium.

Consider an eq. x, and

$$x^0 = \begin{pmatrix} x_1 + \varepsilon_1 \\ x_2 + \varepsilon_2 \\ \vdots \\ x_n + \varepsilon_n \end{pmatrix} \tag{51}$$

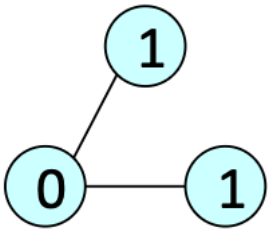
where ε_i is a small perturbation.

Then let x^1 be the best response to x^0, and iterate forward:

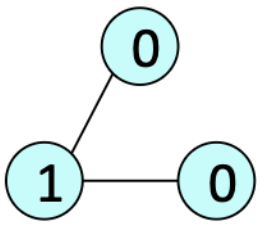
$$x^t = \text{BR}\left(x^{t-1}\right) \tag{52}$$

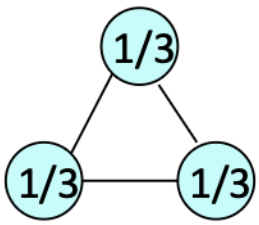
and if after T periods the whole system goes back to x then we say it's stable.

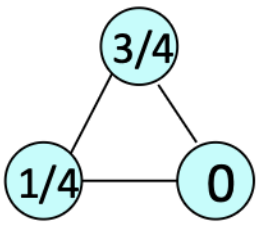
So in the BK model, we can find that the *specialist equilibria* is stable, that is



This equilibrium is stable, whereas the following equilibria







are unstable. The proposition is:

Proposition 3.5.1: Only stable equilibria are specialist equilibria such that every non-specialist has at least two specialists in his or her neighborhood

1. If there is only one specialist, when the specialist's level drops slightly, the free-rider will start providing its own services.
2. If there are at least 2 specialists, even with minor perturbations:
   1. The free-rider still sees a neighbor that is working hard enough.

2. It continues free-riding (remaining at 0).
3. The specialist will also remain at 1 under the best-response condition.
4. The system returns to the equilibrium, so it is stable.

3.5.2. (Jackson, 2019) details

3.5.2.1. Linear-quadratic game

In the society-wide benchmark, agents best respond to a population average ((Jackson, 2019), §3.1.1). Individual expected utility is:

$$\mathbb{E}[U_i^{\text{soc}}(x_i, x_{-i})] = \theta_i x_i + a x_i d_i \mathbb{E}\left[x_j\right] - \frac{c}{2} x_i^2 + \varphi \mathbb{E}\left[\sum_{j \neq i} x_j\right] \tag{53}$$

Its components are:
1. $\theta_i x_i$: the intrinsic preference
2. $a x_i d_i \mathbb{E}\left[x_j\right]$: complementarity with the population mean action.
3. $-\frac{c}{2} x_i^2$: quadratic cost
4. $\varphi \mathbb{E}\left[\sum_{j \neq i} x_j\right]$: a global externality from others' actions. It can be positive or negative and is constant with respect to x_i in an individual's best response. The benchmark is a noncooperative game; a planner would also internalize the effects of x_i on other agents.

Perceived expected utility is:

$$\begin{aligned} \mathbb{E}\left[U_i^{\text{prec}}(x_i, x_{-i})\right] &= \theta_i x_i + a x_i d_i \hat{\mathbb{E}}\left[x_j\right] - \frac{c}{2} x_i^2 + \varphi \mathbb{E}\left[\sum_{j \neq i} x_j\right] \\ &= \mathbb{E}[U_i^{\text{soc}}(x_i, x_{-i})] + a x_i d_i \left(\hat{\mathbb{E}}\left[x_j\right] - \mathbb{E}\left[x_j\right]\right) \end{aligned} \tag{54}$$

Naive agents use their friends' actions to forecast population behavior.

And each agent i solves:

$$\max_{x_i} \quad \theta_i x_i + a x_i d_i \hat{\mathbb{E}}\left[x_j\right] - \frac{c}{2} x_i^2 \tag{55}$$

and the best response is:

$$x_i^* = \frac{1}{c}\left(\theta_i + a d_i \hat{\mathbb{E}}\left[x_j\right]\right) \tag{56}$$

which gives Equation 49.

Lemma 3.5.1: Assume $a, c > 0$, nonnegative types, and common beliefs over the joint distribution of types and degrees. Thus $\hat{\mathbb{E}}_i[d] = \hat{\mathbb{E}}[d]$ and $\hat{\mathbb{E}}_i[\theta] = \hat{\mathbb{E}}[\theta]$ for every agent. If $c > a\hat{\mathbb{E}}[d]$, then there exists a unique eq.:

$$x_i^{\text{prec}}(\theta_i, d_i) = \frac{\theta_i}{c} + \frac{ad_i\hat{\mathbb{E}}[\theta]}{c^2 - ca\hat{\mathbb{E}}[d]} \tag{57}$$

To derive Lemma 3.5.1, start from the best response in Equation 56 and take expectations:

$$\begin{aligned} x_i &= \frac{1}{c}(\theta_i + ad_i\hat{\mathbb{E}}[x]) \\ \hat{\mathbb{E}}[x] &= \hat{\mathbb{E}}\left[\frac{1}{c}(\theta_i + ad_i\hat{\mathbb{E}}[x])\right] \\ \hat{\mathbb{E}}[x] &= \frac{1}{c}\left[\hat{\mathbb{E}}[\theta] + a\hat{\mathbb{E}}[d]\hat{\mathbb{E}}[x]\right] \end{aligned} \tag{58}$$

then we can get $\hat{\mathbb{E}}[x]$ and then plug it into the Equation 56 to get Equation 57.

So it's *higher actions for higher types, higher degree*. Similarly, we have

$$x_i^{\text{soc}}(\theta_i, d_i) = \frac{\theta_i}{c} + \frac{ad_i\mathbb{E}[\theta]}{c^2 - ca\mathbb{E}[d]} \tag{59}$$

Under the equal-mean type assumption of Proposition 3.4.1, Lemma 3.4.1 gives Equation 49.

The degree distributions can also be compared by stochastic dominance.

Definition 3.5.1 (First-Order Stochastic Dominance): Let the CDF of X and Y be $F_X(x) = \Pr\{X \le x\}$ and $F_{Y(y)} = \Pr\{Y \le y\}$, then we say $X \underset{\text{FOSD}}{>} Y$ iff

$$F_X(x) \le F_Y(x), \forall x \in \mathbb{X} \tag{60}$$

and $<$ holds strictly for some x.

When $\mathbb{V}_i[d] > 0$, degree-weighted sampling gives strict first-order stochastic dominance:

$$\hat{P}_i(d) \equiv \frac{P_i(d)d}{\mathbb{E}_i[d]} \underset{\text{FOSD}}{>} P_i(d), \tag{61}$$

and therefore $\hat{\mathbb{E}}_i[d] > \mathbb{E}_i[d]$. If $\mathbb{V}_i[d] = 0$, the two distributions and their means coincide.

4. Cooperation on networks

4.1. Sociology of closure (Coleman, 1988)

When people in a social network know each other (closed structure), it is easier for them to form common norms, build trust, and restrain an individual's bad behavior through collective pressure.

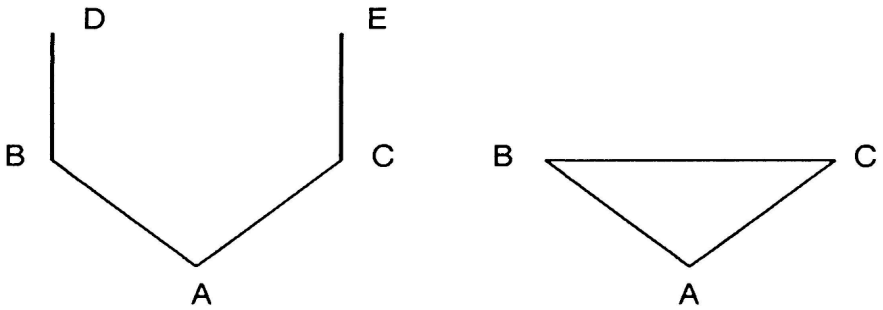


FIG. 1.—Network without (*a*) and with (*b*) closure

Definition 4.1.1 (Collective sanction): Collective sanction refers to the collective pressure, punishment, or constraint exerted on an individual by members of a group in order to establish norms.

4.1.1. Clustering coefficients

1. Prison friendships
   - .31 (MacRae 60) vs .0134
2. Co-authorships
   - .15 math (Grossman 02) vs .00002,
   - .09 biology (Newman 01) vs .00001,
   - .19 econ (Goyal, van der Leij, Moraga 06) vs .00002,
3. Web
   - .11 for web links (Adamic 99) vs .0002

4.2. Cooperation incentives on networks

The game is called *Favor exchange*

1. cost of giving a favor: c
2. net benefit per period is v
3. total net benefit with discount factor δ:

$$V = \sum_{t=1}^{\infty} \delta^t v = \frac{\delta}{1-\delta} v \tag{62}$$

Self-enforcing This refers to cooperation that can be maintained solely through mutual interests, without the need for police or laws.

The incentive condition is:

$$c < V = \frac{\delta}{1-\delta} v \tag{63}$$

4.2.1. Community enforcement

So if it's a one to one relation ship, we need $c < V = \frac{\delta}{1-\delta} v$,but if we have a network like this:

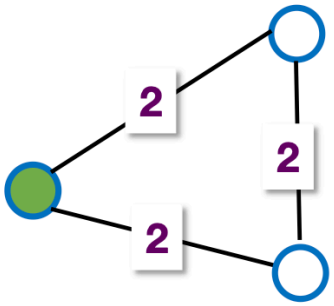


Figure 16: Community-enforcement

In Figure 16, we have one mechanism: **Ostracize deviating agent**

If the green dot does something bad (such as refusing to help one of the white dots). Because the two white dots are friends, they share information. If one of the green dots betrays the other, **both white dots will punish the green dot simultaneously** (for example, by severing their relationship with it).

So in this case, we have

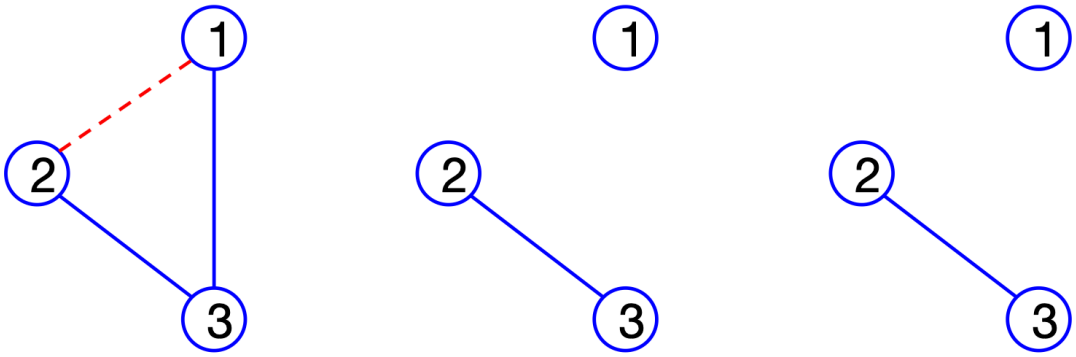
$$c < 2 \times V = 2 \times \frac{\delta}{1-\delta} v \tag{64}$$

4.3. (Jackson et al., 2012)

Ostracism without cost?

If a traitor is kicked out as punishment, the remaining group may be **too small to maintain cooperation**, ultimately causing the entire network to collapse.

Suppose $V < c < 2V$:



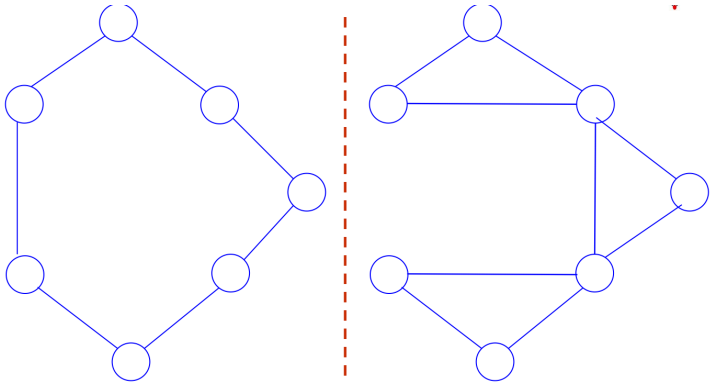
In this case, first 1 deviates against 2, then *1 is ostracized*. But when 3 deviates against 1, she will find that maintain the relationship with 2 is no longer profitable, so the entire network collapses.

Therefore, this threat of punishment is non-credible.

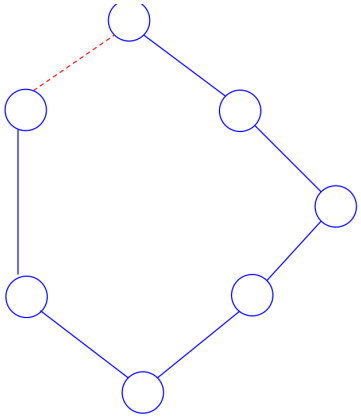
Formally, at period t:

1. At most one agent i_t is called upon to perform a favor for $j_t \in N_i(g_t)$
2. i_t keeps or deletes the link
3. others can respond: announce which (remaining) links they wish to maintain
4. links are retained iff mutually agree
5. the resulting network is g_{t+1}

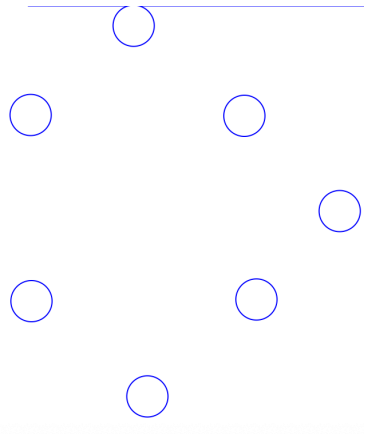
We consider 2 structures:



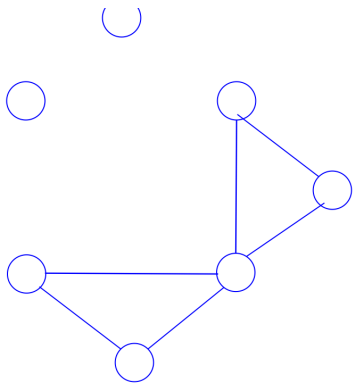
For the left structure, if the there is one disconnection:



The disconnection eventually isolates all nodes:



that is, **autarchy**. But for the right structure, suppose the same split happens, it will end up as:



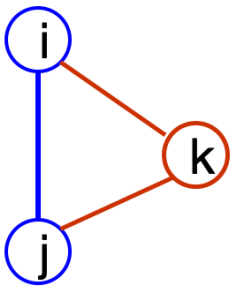
It loses a triad, preserving more links than the left network's autarchy outcome.

Definition 4.3.1 (Robustness Against Contagion): A network such that the punishment for failing to perform a favor only impacts neighbors of original players lose links

Deleting a node or experiencing a disturbance (default) has a *localized* impact.

Definition 4.3.2 (Supported Link): Link $ij \in g$ is supported if there exists k such that $ik \in g, jk \in g$.

That is, i, j, k forms a triangle.



The following theorem gives a necessary condition for robustness:

Theorem 4.3.1: If no pair of players could sustain favor exchange in isolation and a network is robust, then all of its links are supported.

4.4. Closure and support

1. Closure (Coleman, 1988)
2. Support (Jackson et al., 2012)

For closure, we need to ask a question as in Figure 22, that is w/ what frequency are a typical node A's neighbors, say B and C, neighbors of each other?

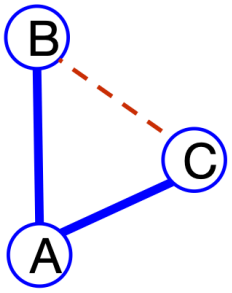


Figure 22: closure

And for support, as in Figure 23, we wonder w/ what frequency do a typical pair of connected nodes, say A and B, have a common neighbor C?

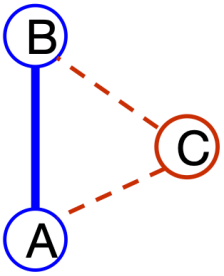


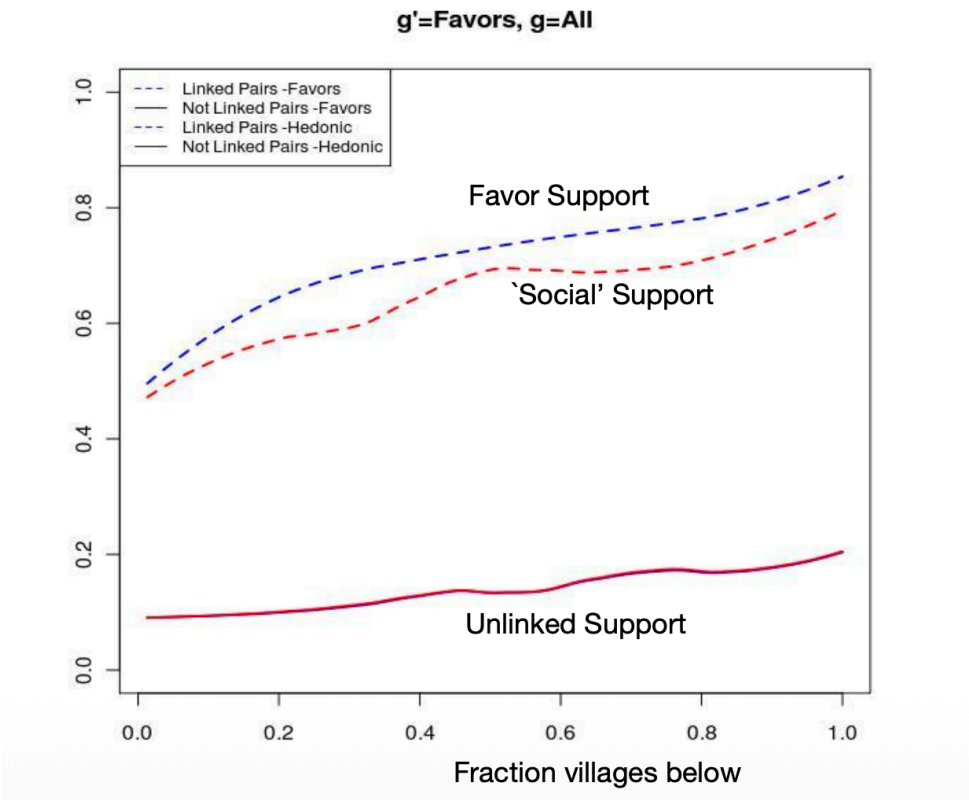
Figure 23: Support

4.5. Evidence from Indian villages

The survey covers 75 rural villages in Karnataka, India. Favor links include lending money or rice and giving advice; social links include visiting and chatting. These data are used in the support analysis (Jackson et al., 2012) and the microfinance study (Banerjee et al., 2013).



The support curves give the fraction of observed links whose endpoints share a neighbor in the broader network. Each position on the horizontal axis is a village. Support varies across villages and is higher than clustering throughout this figure. A favor link can be supported by a common neighbor with whom the endpoints have other kinds of relationships.



Lending links are more common between people with mutual friends. If shared neighbors help enforce repayment, borrowers and lenders have an incentive to form supported links.

4.6. Multiplexity

Multiplex to better sustain cooperation

- prisoner's dilemma: one may deviate against a partner
  - like not return money borrowed
- incentives to prevent such deviations:
  - potential interactions in future on the same link
  - punishments from other (common) friends
- **multiplex**: deviation on one relation is punished by other relations

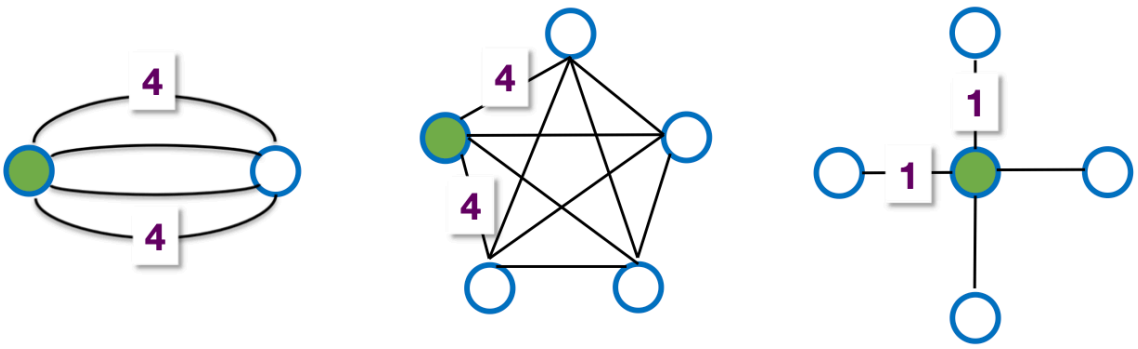


Figure 26: 3 types of networks

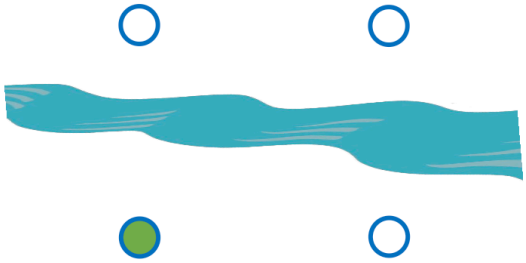
multiplex and community enforcement both enhance cooperation

- multiplex is more robust: rely less on the rest of the network

In Figure 26, we have
- left: multiplex
- middle: community enforcement
- right: star network w/o support

4.6.1. Multiplex trap and homophily

Consider the following network:



so there are 2 types of relations:
1. kid care: better on one side
2. risk share: better across the river

For the green node, she will link kid care w/ the lower right node
1. then she'll decide the risk-share link
2. it's better to link the upper left one
3. but w/ multiplexity, she'll feel it better to link w/ the same lower right node:

In the end, we have the eq. as in the left of Figure 28.
- homophily expands from one relation to other relations
- so called. **multiplex trap**

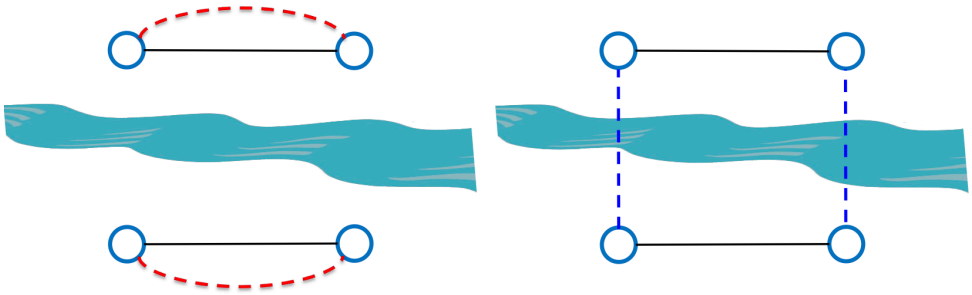


Figure 28: Eq. and efficiency

The most efficient outcome on the right of Figure 28 is not an equilibrium.

5. Financial networks

5.1. (Allen & Gale, 2000)

- Viewed as a starting point
- An arrow: one holds deposits in another bank

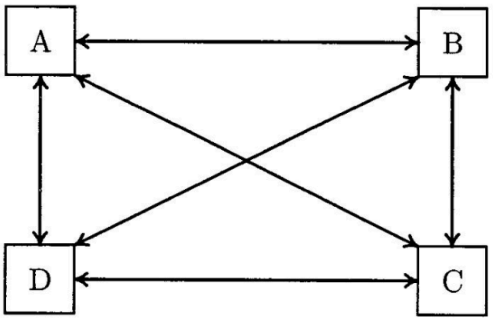


FIG. 1.—Complete market structure

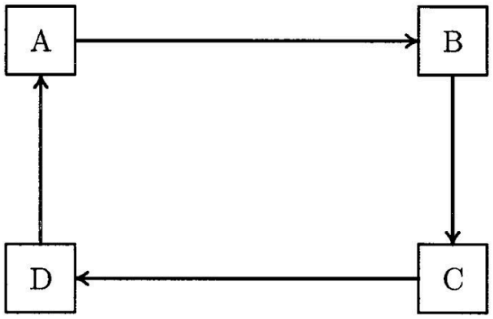


FIG. 2.—Incomplete market structure

The left one is viewed as "complete" and the right one is "incomplete".

Consider a shock to A and makes it bankrupt:

- in the right figure:

$$D \to C \to B \tag{65}$$

- in the left figure: equally shared by $B, C, D \implies$ More likely they all survive.

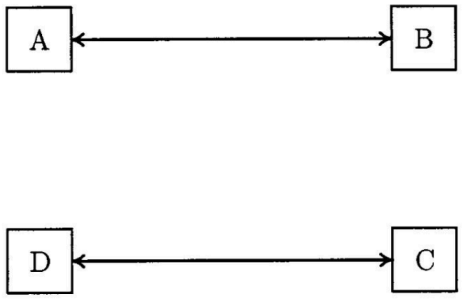


FIG. 3.—Disconnected incomplete market structure

Consider a shock that makes A bankrupt:

- only affect B
- less fragile than fig 2

5.2. (Acemoglu et al., 2015)

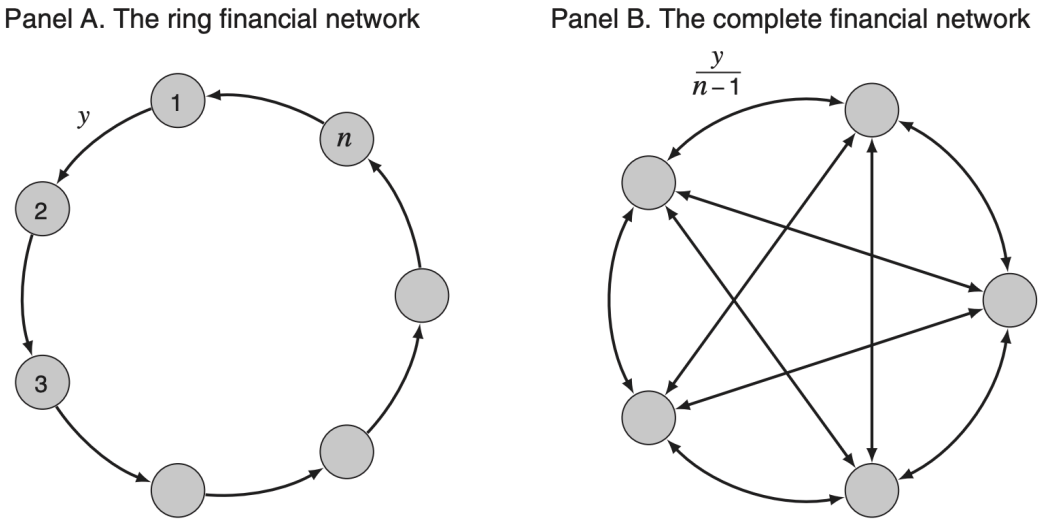


FIGURE 1. THE RING AND THE COMPLETE FINANCIAL NETWORKS

- Arrow: debt contracts

For a **small** negative shock, the ring (left one) is fragile.
- more like a ring: more fragile
- more complete: more stable

For a **large** negative shock:
- **Ring and complete networks are the least stable ones**

but once we have a separation "δ-connected":
- when δ is small, this is more stable than ring or complete.

Phase Transition: under small shocks vs. large shocks, the more stable network structures can be very different

- Pros:
  - generalization to (Allen & Gale, 2000)
  - debt contracts: widely accepted by macro/finance audience
- Cons:
  - model not very tractable
  - limited lessons regarding network structures

5.3. (Elliott et al., 2014)

- develop network-based measures of cascades
- distinguish the effects of *diversification* and *integration*.
- show non-monotonic effects of diversification and integration on contagion.

5.3.1. Basics

- organizations have claims on
  - fundamental assets
  - other organizations
- Bankruptcy cost:
  - when an organization's value falls below a level, the value of other's claims on its drop discontinuously
  - drop in value of one leads to drop in values of others: **cascades**

5.3.2. Model

- organizations: $\{1, 2, ..., n\}$
- Assets: $\{1, 2, ..., m\}$
- let price of asset k be p_k
- and let D_{ik} be the holdings of asset k by organization i

Value of an organization:

- book value:

$$V_i = \sum_{k=1}^{m} D_{ik} p_k + \sum_{j=1}^{n} C_{ij} V_j \tag{66}$$

  - where $\sum_{k=1}^{m} D_{ik} p_k$ is direct asset holdings
  - and $\sum_{j=1}^{n} C_{ij} V_j$ is cross holdings

In vector form:

$$V = Dp + CV \tag{67}$$

that is,

$$V = (I - C)^{-1} Dp \tag{68}$$

- Market Value:
  - defined as value to private investors
  - let

$$v_i = \hat{C}_{ii} V_i \tag{69}$$

  - that is:

$$v = \underbrace{\hat{C}(I - C)^{-1}}_{A} Dp \tag{70}$$

  - and then A_{ij} denotes: *fraction of the returns owned by org j that ultimately accrue to private shareholders of i*

Example 5.3.1: Let there be $n = 2$ firms, each owns half of the other:

$$C = \begin{pmatrix} 0 & 0.5 \\ 0.5 & 0 \end{pmatrix} \tag{71}$$

then it must own half of itself:

$$\hat{C} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \tag{72}$$

and then

$$\begin{aligned} A = \hat{C}(I - C)^{-1} &= \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} \frac{4}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{4}{3} \end{pmatrix} \\ &= \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} \end{aligned} \tag{73}$$

and this is depicted as Figure 30

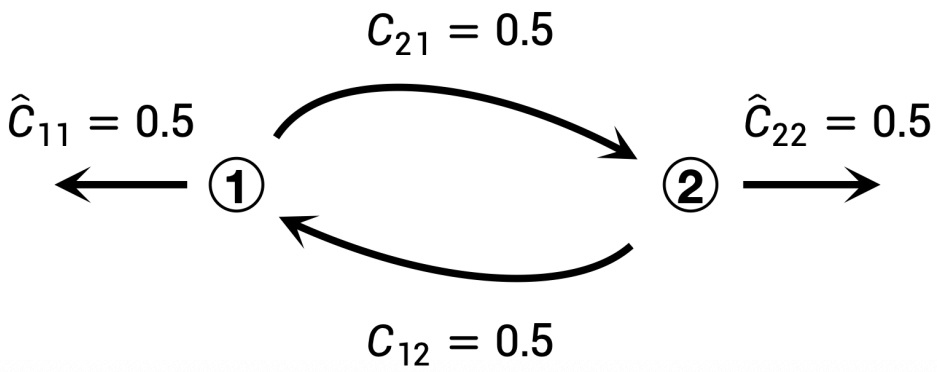


Figure 30: cascade of 2 firms

5.3.3. Bankruptcy

if an org's value drops below some *threshold* θ:

- its value falls discontinuously by β_i
- let the threshold be $\underline{v}_i$, then

$$b_i(v_i) = \begin{cases} \beta_i & \text{if } v_i < \underline{v}_i \\ 0 & \text{o.w.} \end{cases} \tag{74}$$

w/ bankruptcy, we have

$$v = A[Dp - b(v)] \tag{75}$$

Solution

- let $s_i = 1$ if an org. fails
  - $s_i = 0$ if not
- $s = (s_1, ..., s_n)$ is an equilibrium if

$$v = A[Dp - \text{diag}(\beta)s] \tag{76}$$

and

$$v_i < \underline{v}_i \iff s_i = 1 \tag{77}$$

that is

$$s(v) = \mathbb{I}(v < \underline{v}) \in \mathbb{R}^{n \times 1} \tag{78}$$

Equilibria

- Bankruptcies are complements:
  - one fails → others likely to fail
  - the set of equilibria forms a complete lattice
- exists a "base case" eq.
  - fewest orgs fail
- algorithm to find it:
  - identify orgs that fail even all others success
  - then identify failures due to the failure of the first org.

simulation

- one asset per org starting at value 1
- shock is to pick one org.'s direct investment to devalue to 0
- then check if it fails according to Section 5.3.3
- look at the result cascade

Let there be $n = 100$ org. and form a random network w/

$$\Pr\{g_{ij} = 1\} = \frac{d}{n-1} \tag{79}$$

- where d is the expected number of other org. that an org. cross holds
- and d is therefore *level of diversification*

Then we let c fraction of org cross-hold:

- so $1 - c$ holds privately

Then claim i has on j:

$$C_{ij} = c\frac{g_{ij}}{d_j} \tag{80}$$

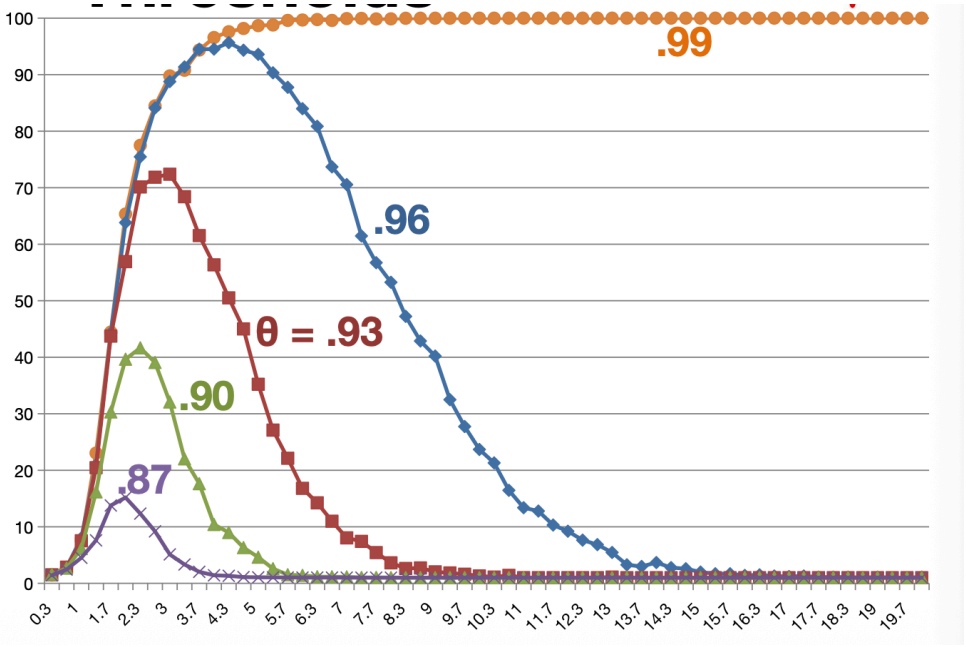


Figure 31: diversification and contagion

The x-axis of Figure 31 is the degree and the y-axis of it is the percent of orgs that fail. different lines reports different threshold θ.

A cascade

Three necessary components of it:
1. a first failure
2. contagion: that is, other org. must be exposed to the failed org. sufficiently to fail too
3. wider propagation: sufficiently large components

Diversification

- has non-monotonicity
  - increases component size
  - decrease the impact of one organization on its neighbors

Intermediate diversification can produce widespread contagion:
- low diversification:
  - fragmented network
  - no widespread contagion
- high
  - little exposure to any single other org.
  - failures do not spread

however, in the middle:
- connected, contagion is possible
- exposure to only a few others makes it easy to spread.

Integration

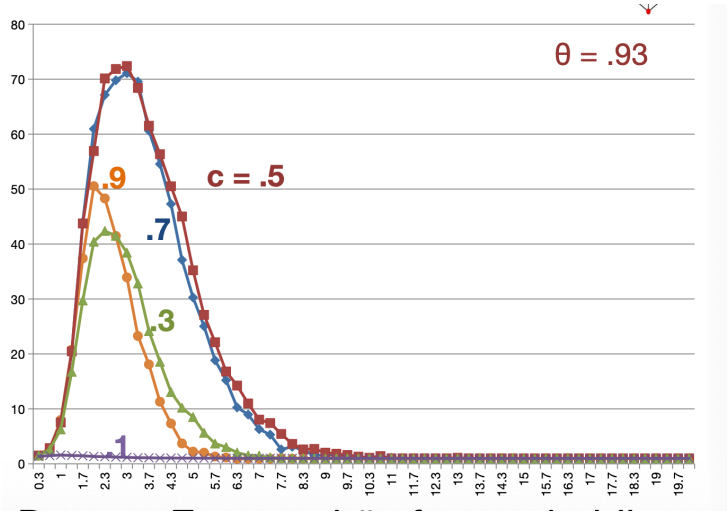


Figure 32: integration

Here the x-axis is the degree: expected number of cross holdings.

- Low:
  - little exposure to others,
  - failures don't trigger others
- high:
  - difficult to get a first failure
  - failure of own assets doesn't trigger failure
- middle:
  - exposure to others enough to trigger contagion

5.3.4. summary

- a first failure:
  - some org. needs to fail
  - *integration decreases own-asset dependence*
- initial contagion:
  - some neighbors need to be sufficiently exposed to fail too

  1. *integration increases exposure of neighbors*
  2. *diversification decreases exposure of specific neighbors*
- Interconnection:
  - to continue to cascade widely, the network must have sufficiently large components.

5.4. Eurodebt (Elliott et al., 2014)

- 6 key countries in Europe that have large cross holdings of each other's sovereign debt
- isolated system
- what happens if values fall and debt is devalued

- Pros:
  - tractable
  - can be taken to data
- Cons
  - cross-holding based model
  - mechanical propagation through cross-holdings, without economic incentives

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